Tristan Cravello

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https://triscravello.github.io/tristan-portfolio-site/

SUMMARY

Software Engineer with hands-on experience building full-stack applications, backend APIs, and distributed systems using **Java, Spring Boot, Python, TypeScript, React, Next.js, and PostgreSQL**. Experienced in concurrency control, database design, automated testing, cloud deployment, and AI-assisted engineering workflows. Former educator bringing strong communication, technical problem-solving, and a focus on building reliable, user-centered software.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
Application & Backend Development: Spring Boot, Spring Data JPA, FastAPI, Node.js, Express, REST APIs
Databases & Data: PostgreSQL, MongoDB, Redis, PostGIS, Prisma, Relational Database Design
Testing & Troubleshooting: JUnit, Pytest, Testcontainers, Debugging, API Testing, OpenAPI
Cloud & Development Tools: AWS, Docker, GitHub Actions, CI/CD, CloudWatch, Git
Frontend: React, Next.js, Tailwind CSS
Additional Technologies: Kafka, Kubernetes, Terraform, Amazon Lambda, Amazon ECS/Fargate, Prometheus, Grafana, Kibana

FEATURED PROJECTS

Distributed Ticket Reservation API | **Distributed Backend Engineering**
Java • Spring Boot • PostgreSQL • Redis • Lua • Docker • Testcontainers • k6

- Built a distributed ticket reservation API with two **Spring Boot** instances, coordinating concurrent booking requests through **PostgreSQL** transactions and Redis admission control.
- Implemented idempotency keys, pessimistic row locking, and database uniqueness constraints to prevent duplicate seat reservations and ensure transactional correctness.
- Developed atomic **Redis Lua** admission leases with configurable fail-open and fail-closed policies to coordinate request concurrency across multiple instances.
- Designed multi-instance concurrency tests, Redis outage simulations, and k6 load benchmarks to validate booking correctness, fault tolerance, and system performance.

Threat Event Platform | **Cloud-Native Event Processing Platform**
Java • Spring Boot • Kafka • Elasticsearch • AWS ECS Fargate • Docker • GitHub Actions

- Built a cloud-native threat event platform using **Java, Spring Boot, PostgreSQL, Kafka, and Docker** to ingest, process, and analyze security events through REST APIs.
- Integrated **Elasticsearch indexing** and Kibana dashboards to enable full-text threat search, filtering, and investigation workflows.
- Deployed to **AWS ECS Fargate** through **GitHub Actions OIDC**, Amazon ECR, Application Load Balancer, CloudWatch, and automated deployment rollback.

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RunLayer | Full-Stack Recommendation Platform
Next.js • TypeScript • PostgreSQL • Prisma • Tailwind CSS

- Built a modular recommendation engine combining weather, user preferences, and scoring algorithms to generate personalized running outfit recommendations.
- Designed normalized **PostgreSQL** schemas supporting dozens of gear items, recommendation history, user profiles, and administrative workflows.
- Implemented comprehensive testing for recommendation logic, category diversification, personalization, and filtering behavior.

Policy-Aware AI Repair Controller | AI Engineering
Python • Claude Code • Pytest • AI Agents

- Built an AI-assisted repair workflow that diagnoses failures and proposes bounded code changes.
- Separated AI-generated repair decisions from deterministic policy enforcement.
- Restricted repair actions to explicitly permitted scopes and used automated tests as acceptance gates.

PROFESSIONAL EXPERIENCE

Springboard, Remote **June 2025 - Present**
Software Engineering Fellow

- Completed an intensive software engineering program focused on full-stack JavaScript development, software architecture, testing, databases, and API development.
- Built and deployed 15+ full-stack, cloud-native applications using **React, Next.js, Node.js, MongoDB, and PostgreSQL.**
- Collaborated on project planning, code reviews, debugging, and iterative feature development across full-stack applications.
- Applied modern development practices, including Git workflows and continuous improvement.

Discovery Academy of Science & Tinker K-8 School **July 2022 - May 2025**
Middle School Social Studies Teacher

- Designed structured instructional systems and assessment workflows for 6th-8th grade World History, Civics, and United States History, serving **10-25** students across multiple learning tracks.
- Applied data-driven evaluation methods to monitor student performance trends and optimize instructional outcomes.

EDUCATION

University of South Florida, Remote **March 2026**
Software Engineering Bootcamp Certification

Florida State University, Tallahassee, Florida **April 2022**
Master's & Bachelor of Science in Social Science Education